Housing Reallocation and Demographic Transitions

A Two-Generation OLG Model

1 Analytical Model

This section studies two implications of housing access in a two-period overlapping-generations economy. A financing-constrained young buyer can value family-sized space more than an old incumbent. We also compare two positive steady states and compute a terminally closed adjustment after a permanent change in desired fertility. In the numerical construction, that adjustment generates house-price appreciation and taxable gains even though gains vanish at either endpoint. Appendix A gives the household solutions, transition results, and mixed-tenure construction.

1.1 Environment and household choices

At date t, the economy contains Y_t young households and O_t old households. A young household has income y_i, liquid wealth b_i, and chooses total nondurable expenditure c, total housing h, completed fertility n, saving a', and tenure $m \in \{R, O\}$. The pair (y_i, b_i) is drawn from a fixed distribution F in each cohort; estates are warm-glow expenditures rather than the source of the next cohort's b_i.

Each child uses $\chi > 0$ units of goods and $\kappa > 0$ units of housing. Define the parts of the household bundle available to the adults as

$$x = c - \chi n > 0, \qquad s = h - \kappa n > 0. \tag{1}$$

Young and old flow utilities are

$$u_t^Y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n, \tag{2}$$

$$u^2(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e. \tag{3}$$

Here e is a warm-glow estate. Writing both c and h as gross choices makes the two child requirements parallel: both reduce the household bundle left for the adults. Logarithmic utility keeps conditional demands and the response to a binding housing limit in closed form; the mechanisms themselves enter through access constraints and tax timing.

A child born at t produces $\nu > 0$ young households at $t+1$, after survival and household formation. Thus $Y_{t+1} = \nu \bar{n}_t Y_t$ and $O_{t+1} = Y_t$, where $\bar{n}_t$ is average fertility among the young.

The fixed housing stock is $\bar{H}$. Rental units satisfy $h \leq h_R^{\max}$, owner units satisfy $h \leq h_O^{\max}$, and $h_R^{\max} < h_O^{\max}$. The consumption good is the numeraire, and the economy can trade it through a one-period bond with price $q \in (0, 1)$ and gross return $R_f = q^{-1}$; only the housing market must clear domestically. The property tax applies to every housing unit and is paid by the owner of record. Owner-occupiers pay it directly; for a rental unit, the landlord is a competitive intermediary that can trade the bond. The intermediary buys one unit for P_t at the beginning of date t. At the end of the period it collects rent r_t, pays property tax $\tau^p P_t$, and carries a unit worth P_{t+1}. Since $R_f = q^{-1}$ is the gross return between the beginning and end of the period, free entry imposes

$$r_t + P_{t+1} = R_f P_t + \tau^p P_t \quad \Longleftrightarrow \quad r_t = (R_f + \tau^p)P_t - P_{t+1}. \tag{4}$$

Thus r_t is paid in date-$(t+1)$ goods and $q\, r_t$ is its date-t value. Household budgets are written in beginning-of-period goods, so rent and the property tax appear at their discounted values. At a steady state, $r = (R_f - 1 + \tau^p)P$. Capital-gains taxation enters through owner-occupied housing. Rental intermediaries are not taxed on resale in the baseline, so the gains tax does not enter (4). Appendix A.5 gives the corresponding incidence calculation when intermediaries are also taxed.

Household capital gains are taxed only upon sale. An old owner who bought one unit at P_{t-1} and sells it at P_t pays the tax per unit

$$\ell_t = \tau^g \max\{P_t - P_{t-1}, 0\}, \tag{5}$$

where τ^g is the capital-gains tax rate. The owner pays tax only when $P_t > P_{t-1}$. If the owner keeps the house until death, appreciation accrued during the owner's life is not taxed. The estate's tax basis is reset to the date-$(t+1)$ price, so an immediate estate sale realizes no taxable gain. The estate sale returns the unit to the fixed stock, and its proceeds enter the warm-glow estate and leave the economy. Selling while alive therefore triggers a tax that retaining the unit avoids. A young buyer finances the share $\phi \in (0,1)$ and therefore posts $(1-\phi)P_t h$ at purchase.

Conditional on renting, the young household solves

$$W_t^R(i) = \max_{c,h,n,a'} u_t^Y(c,h,n) + \beta V_{t+1}^R(R_f a') \tag{6}$$

$$\text{s.t.} \quad c + a' + q\, r_t h = y_i + b_i + T_t, \qquad h \le h_R^{\max}, \qquad a' \ge 0. \tag{7}$$

The renter pays for housing services each period and enters old age with financial wealth $R_f a'$. Conditional on owning, the young household solves

$$\begin{aligned} W_t^O(i) = \max_{c,h,n,a'} \; & u_t^Y(c,h,n) + \beta V_{t+1}^O(a_{t+1}, h) \\ \text{s.t.} \quad & c + a' + [(1-\phi) + q\tau^p]P_t h = y_i + b_i + T_t, \\ & a_{t+1} = R_f a' - R_f \phi P_t h, \qquad (1-\phi)P_t h \le b_i, \qquad h \le h_O^{\max}, \qquad a' \ge 0. \end{aligned} \tag{8}$$

The closing occurs before current income and tax rebates are available, and unsecured borrowing is unavailable. Thus only beginning-of-period liquid wealth b_i can satisfy the down payment. The transfer T_t is the date-t present value of the equal per-household rebate of housing-tax revenue.

An old household has financial wealth a. An owner also carries housing H from young age. Renters and owners solve, respectively,

$$V_t^R(a) = \max_{c^2,h^2,e} u^2(c^2,h^2,e), \qquad c^2 + qe + q\, r_t h^2 = a + T_t, \quad 0 < h^2 \le h_R^{\max}, \tag{9}$$

$$\begin{aligned} V_t^O(a,H) = \max_{c^2,h^2,e} u^2(c^2,h^2,e), \qquad & c^2 + qe + (q\, r_t - \ell_t)h^2 = a + (P_t - \ell_t)H + T_t, \\ & 0 < h^2 \le H, \quad e \ge P_{t+1}h^2. \end{aligned} \tag{10}$$

The marginal cost of retaining a unit is therefore $q\, r_t - \ell_t$: selling realizes the current tax, while retaining the unit avoids it.

Each household draws once an idiosyncratic preference ξ_i for ownership, independently of (y_i, b_i), and observes it before choosing tenure. It owns when $W_t^O(i) + \xi_i \geq W_t^R(i)$. Assume ξ_i is logistic with location $\bar{\xi}$ and scale $\sigma_\xi > 0$, which gives the owner share

$$\pi_t^O(i) = \frac{\exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}{\exp\{W_t^R(i)/\sigma_\xi\} + \exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}. \tag{11}$$

This cross-sectional preference heterogeneity makes aggregate tenure shares smooth. It does not enter the conditional renter or owner policies.

1.2 Housing demand and access

We first ask how much housing a young household would choose if neither the rental limit nor the down-payment constraint restricted it. We then compare that desired amount with the housing the household can actually obtain. If desired housing exceeds the accessible amount, the constraint binds. We then measure the constraint by the household's willingness to pay for one more unit of housing.

Fix a household i, a date t, and a tenure choice $m \in \{R, O\}$. To obtain closed forms, first suppose that saving and the household's old-age choices are interior. Let $A = 1 + \gamma + \omega_B$ denote the sum of the old-age expenditure weights, and let lifetime resources, valued at date t, be

$$w_{it} = y_i + b_i + T_t + qT_{t+1}. \tag{12}$$

The current cost of housing depends on tenure:

$$u_t^R = q\, r_t, \qquad u_t^O = q(r_t + \ell_{t+1}). \tag{13}$$

A renter pays the present value of rent. A buyer faces the same opportunity cost and, if the unit is sold in old age, the anticipated gains tax. Solving the saving and old-age choices reduces the lifetime problem, up to a constant, to

$$\begin{aligned} \max_{x,s,n} \quad & (1 + \beta A) \log x + \alpha \log s + \vartheta_t \log n \\ \text{s.t.} \quad & (1 + \beta A)x + u_t^m s + (\chi + \kappa u_t^m)n = w_{it}. \end{aligned} \tag{14}$$

The term βA summarizes old-age demand for consumption, housing, and the estate. With logarithmic utility, these three uses receive fixed expenditure shares, so the same term appears in the reduced objective and budget. The last term in the budget gives the full resource cost of another child: χ units of goods and κ units of housing valued at the tenure-specific user cost.

Without a housing-access constraint, define $D_t = 1 + \beta A + \alpha + \vartheta_t$. The household then chooses usable goods, usable housing, and fertility according to

$$x^m = \frac{w_{it}}{D_t}, \qquad s^m = \frac{\alpha w_{it}}{D_t u_t^m}, \qquad n^m = \frac{\vartheta_t w_{it}}{D_t(\chi + \kappa u_t^m)}. \tag{15}$$

Gross choices are $c^m = x^m + \chi n^m$ and $h^m = s^m + \kappa n^m$.

Access may prevent the household from choosing this amount of housing. A renter cannot exceed the largest rental unit, while a buyer cannot exceed either the largest owner unit or the amount supported by liquid wealth:

$$\widehat{h}_{it}^R = h_R^{\max}, \qquad \widehat{h}_{it}^O = \min\left\{h_O^{\max}, \frac{b_i}{(1 - \phi)P_t}\right\}. \tag{16}$$

If desired gross housing in (15) is below this limit, the constraint is irrelevant. Otherwise the household sets $h^m = \widehat{h}^m_{it}$, and usable consumption and fertility solve

$$(1+\beta A)x^m + \chi n^m + u^m_t \widehat{h}^m_{it} = w_{it}, \tag{17}$$

$$\frac{\vartheta_t}{n^m} = \frac{\chi}{x^m} + \frac{\alpha\kappa}{\widehat{h}^m_{it} - \kappa n^m}. \tag{18}$$

The second equation prices the goods and housing taken from the adult bundle by another child. The appendix proves that these two equations have a unique interior solution on the stated branch.

The value of relaxing the housing limit is the household's marginal value of space minus its user cost:

$$\zeta^m_{it} = \frac{\alpha x^m}{s^m} - u^m_t \geq 0. \tag{19}$$

It is zero when housing is unconstrained and positive when an access limit binds with a positive multiplier. If the owner's physical size limit is slack and the down-payment constraint binds, denote this financial component by $\zeta^{O,F}_{it}$. Combining the housing and fertility conditions gives

$$\frac{\vartheta_t x^m}{n^m} = \chi + \kappa(u^m_t + \zeta^m_{it}). \tag{20}$$

Thus a binding access constraint raises the effective cost of a child by $\kappa\zeta^m_{it}$. This is the direct link between housing access and fertility. When the rental limit also binds in old age, committed old-age rent changes the lifetime budget. Appendix A.2 gives that branch.

1.3 Positive steady states and transitions

The second exercise asks how the economy responds to a permanent change in desired fertility. We first characterize the old and new steady states and then compute a terminally closed approximation to their adjustment. House prices and taxable gains are constant at either endpoint, but can change during the transition.

At an arbitrary date, let $\bar{n}_t$, $\bar{h}^Y_t$, and $\bar{h}^O_t$ denote average fertility, young housing, and old housing generated by current choices and the old-state distribution. At constant prices, transfers, and preferences, write their stationary counterparts as functions of $(P, T; \vartheta)$, and let $\bar{h}(P, T; \vartheta) = \bar{h}^Y(P, T; \vartheta) + \bar{h}^O(P, T; \vartheta)$. Market clearing and the government budget are

$$Y_t \bar{h}^Y_t + O_t \bar{h}^O_t = \bar{H}, \tag{21}$$

$$(Y_t + O_t)T_t = q\tau^p P_t \bar{H} + \ell_t O_t \bar{h}^{\text{sell}}_t, \tag{22}$$

where $\bar{h}^{\text{sell}}_t$ is housing sold by old owners per old household, assigning zero sales to renters.

Proposition 1 (Steady-state accounting)**.** *Every positive steady state satisfies*

$$\bar{n}(P^*, T^*; \vartheta) = \frac{1}{\nu}, \qquad \ell^* = 0, \tag{23}$$

$$T^* = \frac{q\tau^p P^*}{2}\bar{h}(P^*, T^*; \vartheta), \qquad Y^* = O^* = \frac{\bar{H}}{\bar{h}(P^*, T^*; \vartheta)}. \tag{24}$$

Stationarity fixes average fertility at the replacement rate. Prices, transfers, tenure, individual fertility and housing, and the population scale can nevertheless differ across steady states.

Stationarity requires replacement fertility; household choices and housing clearing determine the price and population scale consistent with it. Appendix A.3 gives conditions for local determination.

Suppose ϑ_t rises permanently at date zero. The economy starts from the old steady state, so $S_0 = (Y^0, Y^0, G^0, P^0)$ and $P_{-1} = P^0$. Because the old at date t made their choices when young at $t-1$, the state must keep track of their financial wealth, housing, and tenure. Let G_t denote this distribution and write $S_t = (Y_t, O_t, G_t, P_{t-1})$. Given S_0, a transition equilibrium is a sequence of prices, transfers, household choices, tenure shares, and distributions such that households optimize, the cohort and distribution laws hold, and the following two conditions hold at every date:

$$Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}, \tag{25}$$

$$(Y_t + O_t) T_t = q\tau^p P_t \bar{H} + \ell_t O_t \bar{h}_t^{\text{sell}}. \tag{26}$$

The first equation clears the housing market. The second rebates property-tax revenue and the gains taxes paid by old sellers. The sequence must also converge to the new steady state.

Figure 1 uses a finite terminal closure. From date $J+1$ onward, prices and transfers equal their new steady-state values. The last young cohort faces those values in old age, but its purchase price P_J remains its tax basis, so ℓ_{J+1} can be nonzero. Its induced state is recorded rather than imposed. The computed path satisfies equilibrium through date J and approximates the remaining infinite sequence. Appendix A.4 states conditions for a locally determined solution to each finite system and shows that extending the terminal date barely changes the reported path. These checks do not establish existence or uniqueness of the infinite-horizon transition.

One permanent shock is sufficient for several cohorts to realize gains. The cohort law can be iterated as

$$Y_t = Y_0 \prod_{s=0}^{t-1} \nu \bar{n}_s, \qquad O_t = Y_{t-1}. \tag{27}$$

The first fertility response therefore changes the next young cohort, which later becomes old and produces another cohort. With a fixed housing stock, this sequence can move prices for more than two dates. Whenever $P_s > P_{s-1}$, the old cohort selling at s realizes $\ell_s = \tau^g(P_s - P_{s-1}) > 0$. The preference change is the only exogenous shock in the construction below; the sign and duration of appreciation are properties of its computed equilibrium path.

1.4 Intergenerational reallocation and fertility

Consider moving a small amount of housing from an old incumbent to a financing-constrained young owner. Fertility, cohort sizes, tenure, prices, and all unaffected allocations are held fixed. The old incumbent is interior in both housing retention and estate composition. The young owner saves, has a binding down-payment constraint and a slack physical size limit, and is interior in the relevant old-age choices. Their marginal values of current housing services, expressed in units of current consumption, are

$$MV_{it}^Y = q\, r_t + q\ell_{t+1} + \zeta_{it}^{O,F}, \qquad MV_{jt}^O = q\, r_t - \ell_t. \tag{28}$$

The two households share the housing-service term qr_t. The old owner loses ℓ_t upon sale, the young buyer anticipates $q\ell_{t+1}$, and the binding down-payment constraint contributes $\zeta_{it}^{O,F}$.

To give this comparison a welfare meaning without assigning weights to future people, call an allocation *conditionally locally efficient* if no small reallocation of existing housing, together with budget-balanced transfers, can make the households alive at t weakly better off and at least one

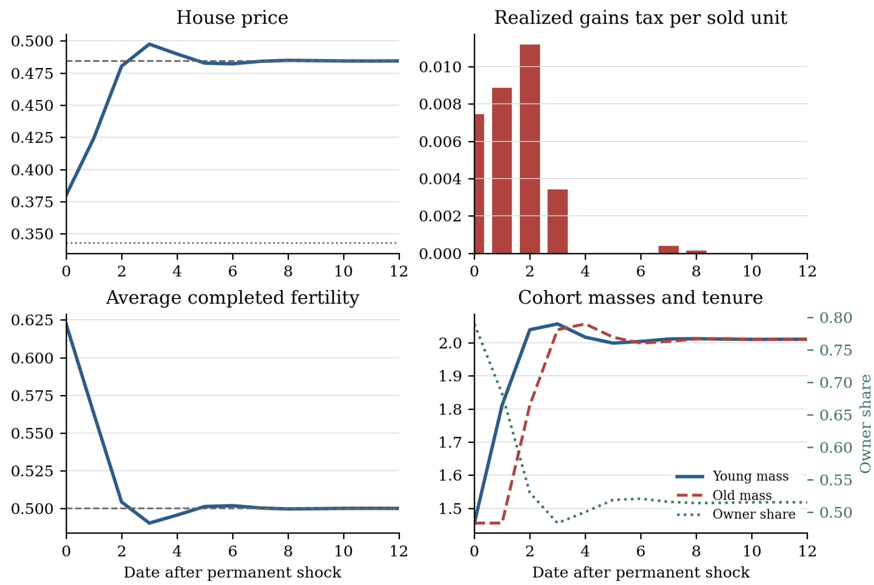

Figure 1: Terminally closed mixed-tenure transition after a permanent increase in the fertility preference. Dotted and dashed lines mark the two steady-state prices; the horizontal fertility line is replacement $1/\nu$. Both tenures have positive mass throughout, and the preference change is the only shock.

strictly better off while fertility, cohort sizes, and tenure are fixed. Suppose an authority can supply pledgeable bridge finance at the bond price and return incremental tax receipts. At $t+1$, it sells the transferred unit while holding the buyer's baseline old-age housing choice fixed. Strict slackness of the buyer's old-age constraints makes this marginal path feasible.

Proposition 2 (Conditional local efficiency test)**.** *Under the conditions above and the two-date transfer ledger stated in Appendix A.5, the marginal-value difference is*

$$MV_{it}^{Y} - MV_{jt}^{O} = \zeta_{it}^{O,F} + \ell_t + q\ell_{t+1}. \tag{29}$$

Let $K_{ijt} \geq 0$ be the marginal real resource cost, in current goods, of transferring title and occupancy from j to i. Purchase and tax payments are transfers, not resource costs. If $MV_{it}^{Y} - MV_{jt}^{O} > K_{ijt}$, the allocation is not conditionally locally efficient.

At a steady state, both tax terms vanish, so the condition is $\zeta_i^{O,F} > K_{ij}$; the tax terms are transition-specific lock-in wedges. The result is a Pareto statement for households alive at t, not a ranking of equilibria with different populations, which requires weights on future people.

The housing constraint can also affect fertility. The sign is not automatic: another child requires both χ units of goods and κ units of space. Holding tenure, prices, lifetime wealth, and the active set fixed, and using the interior old-age branch above, Appendix A.6 shows that

$$\frac{\mathrm{d}n}{\mathrm{d}\widehat{h}} \geq 0 \quad \Longleftrightarrow \quad \chi \leq \frac{(1+\beta A)\kappa(u_t^m + \zeta_{it}^m)^2}{\alpha u_t^m}. \tag{30}$$

More accessible space raises fertility when children are sufficiently space-intensive relative to their goods cost. If the old rental limit also binds, βA is replaced by $\beta(1+\omega_B)$. This partial-equilibrium calculation also gives $\partial n/\partial w > 0$ and $\partial n/\partial u_t^m < 0$. It does not sign a funded policy because prices, rebates, tenure, and lifetime resources also respond.

A Proofs and computational construction

This appendix derives the conditional household policies, gives sufficient conditions for positive steady states and finite-horizon transition truncations, proves the reallocation and fertility results, and reports a reproducible mixed-tenure transition approximation.

A.1 Old-age problems

An old renter with financial wealth a solves

$$V_t^R(a) = \max_{c^2,h^2,e} \; \log c^2 + \gamma \log h^2 + \omega_B \log e \qquad (31)$$
$$\text{s.t.} \quad c^2 + qe + q\, r_t h^2 = a + T_t, \qquad 0 < h^2 \le h_R^{\max}.$$

For an old owner, define

$$A = 1 + \gamma + \omega_B, \qquad \varrho_t = q\, r_t - \ell_t, \qquad X_t = a + (P_t - \ell_t)H + T_t. \qquad (32)$$

The owner problem is (10). Both old problems are strictly concave on their feasible sets.

Lemma 1 (Old-age allocations)**.** *Suppose $r_t > 0$, $\varrho_t > 0$, and resources place the household in the log domain. If the renter size limit is slack, then*

$$c^2 = \frac{a + T_t}{A}, \qquad h^2 = \frac{\gamma(a + T_t)}{Aq\, r_t}, \qquad e = \frac{\omega_B(a + T_t)}{Aq}. \qquad (33)$$

If the renter limit binds, set $Z^R = a + T_t - q\, r_t h_R^{\max}$; then

$$h^2 = h_R^{\max}, \qquad c^2 = \frac{Z^R}{1 + \omega_B}, \qquad e = \frac{\omega_B Z^R}{(1 + \omega_B)q}. \qquad (34)$$

If both owner constraints are slack, then

$$c^2 = \frac{X_t}{A}, \qquad h^2 = \frac{\gamma X_t}{A\varrho_t}, \qquad e = \frac{\omega_B X_t}{Aq}, \qquad (35)$$

and the two strict slackness conditions are

$$\frac{\gamma X_t}{A\varrho_t} < H, \qquad \omega_B \varrho_t > \gamma q P_{t+1}. \qquad (36)$$

Proof. For a generic interior budget $c^2 + qe + p_h h^2 = X$, the three first-order conditions imply $p_h h^2 = \gamma c^2$ and $qe = \omega_B c^2$. The budget gives $Ac^2 = X$. Substituting $p_h = q\, r_t$ for renters and $p_h = \varrho_t$ for owners gives (33) and (35). If renter housing is fixed at $h_R^{\max}$, strict concavity allocates the remaining resources between consumption and the estate in shares $1 : \omega_B$, which gives (34). Substitution of the interior owner policy into $h^2 < H$ and $e > P_{t+1}h^2$ yields (36). □

The remaining owner active sets are also explicit. If only estate composition binds, let $d_t = \varrho_t + qP_{t+1}$; then

$$c^2 = \frac{X_t}{A}, \qquad h^2 = \frac{(\gamma + \omega_B)X_t}{Ad_t}, \qquad e = P_{t+1}h^2. \qquad (37)$$

If only retention binds, set $Z_t = X_t - \varrho_t H$; then

$$h^2 = H, \qquad c^2 = \frac{Z_t}{1+\omega_B}, \qquad e = \frac{\omega_B Z_t}{(1+\omega_B)q}. \tag{38}$$

If both constraints bind,

$$h^2 = H, \qquad e = P_{t+1}H, \qquad c^2 = X_t - (\varrho_t + qP_{t+1})H. \tag{39}$$

The estate-only row requires $\gamma q P_{t+1} \geq \omega_B \varrho_t$ and $h^2 < H$. The retention-only row requires

$$\omega_B Z_t > (1+\omega_B) q P_{t+1} H, \qquad \gamma Z_t \geq (1+\omega_B)\varrho_t H. \tag{40}$$

For the both-binding row, the multipliers are nonnegative exactly when

$$\frac{q}{c^2} \geq \frac{\omega_B}{P_{t+1}H}, \qquad \frac{\gamma + \omega_B}{H} \geq \frac{\varrho_t + qP_{t+1}}{c^2}. \tag{41}$$

These rows and complementary slackness exhaust the strictly concave owner problem.

In the interior owner region, the indirect value and relevant envelopes are

$$V_t^O(a,H) = A\log X_t - A\log A + \gamma \log\gamma - \gamma\log\varrho_t + \omega_B\log\omega_B - \omega_B \log q, \tag{42}$$

$$V_{a,t}^O = \frac{A}{X_t} = \frac{1}{c_t^2}, \qquad V_{H,t}^O = (P_t - \ell_t)V_{a,t}^O. \tag{43}$$

A.2 Young household solutions and tenure

The renter and owner problems are those in the main text. Let λ_t^m be the current-budget multiplier, η_t^m the multiplier on the tenure-specific housing limit, υ_t^m the multiplier on $a' \geq 0$, and μ_t the owner down-payment multiplier. Using $x_t^m = c_t^m - \chi n_t^m$ and $s_t^m = h_t^m - \kappa n_t^m$, the common consumption and fertility conditions are

$$\lambda_t^m = \frac{1}{x_t^m}, \tag{44}$$

$$\frac{\vartheta_t}{n_t^m} = \frac{\chi}{x_t^m} + \frac{\alpha\kappa}{s_t^m}. \tag{45}$$

The saving conditions are

$$\lambda_t^R = \beta R_f V_{a,t+1}^R + \upsilon_t^R, \tag{46}$$

$$\lambda_t^O = \beta R_f V_{a,t+1}^O + \upsilon_t^O, \qquad \upsilon_t^m a_t'^m = 0. \tag{47}$$

The main text presents the interior old-age branch. To include the case in which an old renter is constrained at $h_R^{\max}$, define

$$(k_{t+1}^m, \widetilde{w}_{it}^m) = \begin{cases} (\beta A, w_{it}), & \text{if old-age choices are interior,} \\ (\beta(1+\omega_B), w_{it} - q^2\, r_{t+1} h_R^{\max}), & \text{if } m = R \text{ and the old rental limit binds.} \end{cases} \tag{48}$$

Here k_{t+1}^m is the saving load implied by the Euler equation and $\widetilde{w}_{it}^m$ is lifetime wealth net of any committed old-age rent. On either branch, the young household's reduced lifetime problem, up to a constant, is

$$\max_{x,s,n} \quad (1 + k_{t+1}^m)\log x + \alpha\log s + \vartheta_t \log n$$

$$\text{s.t.}\quad (1+k_{t+1}^m)x + u_t^m s + (\chi + \kappa u_t^m)n = \widetilde{w}_{it}^m. \tag{49}$$

Equivalently, its budget in gross housing is

$$(1+k_{t+1}^m)x + \chi n + u_t^m h = \widetilde{w}_{it}^m. \tag{50}$$

For a renter on either old-age branch in (48), and for an owner whose next-period retention and estate-composition constraints are slack, the housing conditions reduce to

$$\frac{\alpha x_t^m}{s_t^m} = u_t^m + \zeta_t^m, \qquad \zeta_t^R = \frac{\eta_t^R}{\lambda_t^R}, \qquad \zeta_t^O = \frac{\mu_t}{\lambda_t^O}(1-\phi)P_t + \frac{\eta_t^O}{\lambda_t^O}. \tag{51}$$

This proves (19). If the owner-size limit is slack, the second term in ζ_t^O is zero and the access wedge is purely financial.

Writing $B_{it}^m = 1 + k_{t+1}^m + \alpha + \vartheta_t$, an unconstrained household chooses

$$x_t^m = \frac{\widetilde{w}_{it}^m}{B_{it}^m}, \qquad s_t^m = \frac{\alpha \widetilde{w}_{it}^m}{B_{it}^m u_t^m}, \qquad n_t^m = \frac{\vartheta_t \widetilde{w}_{it}^m}{B_{it}^m(\chi + \kappa u_t^m)}. \tag{52}$$

Gross choices are $c_t^m = x_t^m + \chi n_t^m$ and $h_t^m = s_t^m + \kappa n_t^m$. If the current housing limit binds, substituting $h_t^m = \widehat{h}_{it}^m$ into the budget and fertility condition leaves the single equation

$$\frac{\vartheta_t}{n} - \frac{\alpha\kappa}{\widehat{h}_{it}^m - \kappa n} - \frac{(1+k_{t+1}^m)\chi}{\widetilde{w}_{it}^m - u_t^m \widehat{h}_{it}^m - \chi n} = 0. \tag{53}$$

This is the substituted version of the two transparent conditions (17)–(18) in the main text.

Proposition 3 (Lifetime reduction and conditional policies)**.** *Suppose young saving is interior. For owners, suppose next-period old retention and estate composition are interior. For renters, allow either an interior old housing choice or a strictly binding old rental limit. Then the branch objects in* (48) *give the lifetime budget* (50)*. If the current housing limit is slack, the policies are* (52)*. Saving on the interior old-renter, capped old-renter, and interior old-owner branches is, respectively,*

$$a_t'^{R,I} = \beta A x_t^R - qT_{t+1}, \tag{54}$$
$$a_t'^{R,C} = \beta(1+\omega_B)x_t^R - qT_{t+1} + q^2\, r_{t+1} h_R^{\max}, \tag{55}$$
$$a_t'^{O} = \beta A x_t^O - qT_{t+1} - [q(P_{t+1} - \ell_{t+1}) - \phi P_t]h_t^O. \tag{56}$$

The capped renter row applies when its implied old allocation satisfies $\gamma c_{t+1}^2 / h_R^{\max} > q\, r_{t+1}$; the interior row applies with the reverse strict inequality. Equality is the boundary between the two rows. If the effective housing limit binds, $h_t^m = \widehat{h}_{it}^m$ and fertility is the unique root of (53) *in*

$$0 < n < \min\left\{\frac{\widehat{h}_{it}^m}{\kappa}, \frac{\widetilde{w}_{it}^m - u_t^m \widehat{h}_{it}^m}{\chi}\right\}. \tag{57}$$

Proof. On every stated branch, the old value satisfies $V_{a,t+1} = 1/c_{t+1}^2$. The saving Euler equation therefore implies $c_{t+1}^2 = \beta R_f x_t$. For an owner, let

$$X_{t+1} = R_f a_t' - R_f \phi P_t h_t + (P_{t+1} - \ell_{t+1})h_t + T_{t+1}. \tag{58}$$

Interior old choices give $qX_{t+1} = \beta A x_t$. Substitute this expression and the current owner budget into that identity. The mortgage inflow and repayment cancel, leaving $(1+\beta A)x_t + \chi n_t + u_t^O h_t = w_{it}$.

For an interior old renter, $X_{t+1} = R_f a'_t + T_{t+1}$ and the same argument gives $a'_t = \beta A x_t - qT_{t+1}$. If the old rental limit binds, instead define

$$Z^R_{t+1} = R_f a'_t + T_{t+1} - q\, r_{t+1} h^{\max}_R. \tag{59}$$

Equation (34) gives $c^2_{t+1} = Z^R_{t+1}/(1+\omega_B)$, so the Euler equation implies $qZ^R_{t+1} = \beta(1+\omega_B)x_t$. Combining this identity with the young budget gives the second row of (48), the capped saving rule (55), and (50). Thus the committed old-age rent changes both the consumption weight and effective lifetime resources.

When housing is unconstrained, (51) gives $s = \alpha x/u^m_t$ and (45) gives $n = \vartheta_t x/(\chi + \kappa u^m_t)$. Substitution into the lifetime budget gives $\widetilde{w}^m_{it} = (1 + k^m_{t+1} + \alpha + \vartheta_t)x$, proving (52).

When the limit binds, substitute $h = \widehat{h}$ and $x = (\widetilde{w}^m - u^m_t \widehat{h} - \chi n)/(1 + k^m_{t+1})$ into the fertility condition. This gives (53). Its left side tends to $+\infty$ at the lower boundary of (57), tends to $-\infty$ at the upper boundary, and has derivative

$$-\frac{\vartheta_t}{n^2} - \frac{\alpha\kappa^2}{(\widehat{h} - \kappa n)^2} - \frac{(1 + k^m_{t+1})\chi^2}{(\widetilde{w}^m - u^m_t \widehat{h} - \chi n)^2} < 0. \tag{60}$$

The root therefore exists and is unique. $\square$

If saving is zero or a different old-age active set applies, the lifetime reduction is replaced by the KKT system, but strict concavity still makes each conditional tenure policy unique. The paper results state their active-set conditions explicitly rather than extending closed forms through those corners.

With the logistic ownership preference in (11), the conditional probability of ownership is strictly between zero and one whenever the two conditional values are finite. Because the owner-share formula and the conditional policies are continuous, aggregate choices are continuous across the rent–own indifference locus. Other atomless continuous preference distributions give the same regularity, although not generally the same closed-form share.

A.3 Aggregation and positive steady states

Let F denote the distribution of observed young types $i = (y_i, b_i)$. The old-state distribution is over financial wealth a, inherited owner housing H, and tenure. For any bounded measurable test function ψ, its law of motion is

$$\begin{aligned}\int \psi \,\mathrm{d}G_{t+1} = \int \Big[&(1 - \pi^O_t(i))\psi(R_f a'^R_t(i), 0, R) \\ &+ \pi^O_t(i)\psi(R_f a'^O_t(i) - R_f \phi P_t h^O_t(i), h^O_t(i), O)\Big] \,\mathrm{d}F(i).\end{aligned} \tag{61}$$

Average fertility and young housing integrate the corresponding conditional policies with the same weights. Average old housing integrates the old policies over G_t. Average taxable sales are

$$\bar{h}^{\text{sell}}_t = \int \mathbf{1}\{m = O\}[H - h^2_t(a, H, O)] \,\mathrm{d}G_t. \tag{62}$$

Equations (21)–(22), the cohort laws, and (61) close the economy.

Proof of Proposition 1. In a steady state, $O_{t+1} = Y_t$ implies $O^* = Y^*$. Stationarity of the young cohort then implies $1 = \nu \bar{n}(P^*, T^*; \vartheta)$. A constant real price makes $\ell^* = 0$ directly from (5). Housing clearing gives $Y^* = \bar{H}/\bar{h}$. The government budget becomes $2Y^*T^* = q\tau^p P^* \bar{H}$. Eliminating Y^* gives the second equation in (24). Reversing these steps establishes sufficiency once the household policies and stationary distribution are constructed. □

The following result gives checkable sufficient conditions for existence. It also makes clear that a positive steady state is not guaranteed for arbitrary primitives.

Proposition 4 (Steady-state existence and local uniqueness)**.** *Let $\mathcal{D} = [\underline{P}, \overline{P}] \times [\underline{T}, \overline{T}] \subset \mathbb{R}^2_{++}$ be a rectangle on which conditional policies are feasible, relevant user costs are positive, and $\bar{n}$ and $\bar{h}$ are continuous. For the local-uniqueness statement, suppose they are continuously differentiable near the steady state. Define*

$$\Psi(P, T) = \frac{q\tau^p P}{2} \bar{h}(P, T; \vartheta). \tag{63}$$

Suppose:

1. *$\Psi(P, \cdot)$ maps $[\underline{T}, \overline{T}]$ into itself for every P, and there is an $L < 1$ such that $|\Psi(P, T) - \Psi(P, T')| \le L|T - T'|$ for all $P \in [\underline{P}, \overline{P}]$ and $T, T' \in [\underline{T}, \overline{T}]$;*
2. *writing $T = \mathcal{T}(P)$ for its unique fixed point,*

$$[\nu \bar{n}(\underline{P}, \mathcal{T}(\underline{P}); \vartheta) - 1][\nu \bar{n}(\overline{P}, \mathcal{T}(\overline{P}); \vartheta) - 1] < 0. \tag{64}$$

Then a positive steady state exists in $\mathcal{D}$. It is locally unique if

$$J^* = \begin{bmatrix} \nu \bar{n}_P & \nu \bar{n}_T \\ -\dfrac{q\tau^p}{2}(\bar{h} + P\bar{h}_P) & 1 - \dfrac{q\tau^p P}{2} \bar{h}_T \end{bmatrix}_{(P^*, T^*; \vartheta)} \tag{65}$$

is nonsingular.

Proof. The contraction theorem gives a unique fiscal fixed point $\mathcal{T}(P)$ for each P. Uniform contraction and continuity of Ψ imply continuity of $\mathcal{T}$. The intermediate value theorem applied to the replacement residual along this locus gives a price P^* satisfying the first steady-state equation. Its fiscal fixed point $T^* = \mathcal{T}(P^*)$ satisfies the second. Local uniqueness follows from the implicit-function theorem when J^* is nonsingular. □

The first condition can be checked from bounds on primitives and policies. For example, suppose $0 < \underline{h} \le \bar{h} \le \overline{h}$ on $\mathcal{D}$ and $\bar{h}(P, \cdot)$ is uniformly Lipschitz with constant L_h. It is enough to choose transfer bounds such that

$$\underline{T} \le \frac{q\tau^p \underline{P}}{2} \underline{h}, \qquad \overline{T} \ge \frac{q\tau^p \overline{P}}{2} \overline{h}, \qquad \frac{q\tau^p \overline{P}}{2} L_h < 1. \tag{66}$$

The replacement boundary condition says that the low- and high-price faces place average fertility on opposite sides of $1/\nu$ after closing the fiscal account. It is an inequality in the unique household policies, and hence is directly verifiable from primitives rather than an assumed equilibrium.

If J^* is nonsingular, the response to a permanent change in ϑ is

$$\begin{bmatrix} P^*_\vartheta \\ T^*_\vartheta \end{bmatrix} = -(J^*)^{-1} \begin{bmatrix} \nu \bar{n}_\vartheta \\ -\dfrac{q\tau^p P^*}{2} \bar{h}_\vartheta \end{bmatrix}. \tag{67}$$

The associated population-scale response is

$$\frac{\mathrm{d}\log Y^*}{\mathrm{d}\vartheta} = -\frac{\bar{h}_P P^*_\vartheta + \bar{h}_T T^*_\vartheta + \bar{h}_\vartheta}{\bar{h}}. \tag{68}$$

There is no general sign without restrictions on the demand and tenure elasticities.

A.4 Transition equilibrium

Suppose $\vartheta_t = \vartheta^0$ for $t < 0$ and $\vartheta_t = \vartheta^1$ for $t \geq 0$, and both values support regular positive steady states $\mathcal{E}^0$ and $\mathcal{E}^1$. The date-zero state is the old steady state: $S_0 = (Y^0, Y^0, G^0, P^0)$. In particular, $P_{-1} = P^0$ is the basis of date-zero old owners. No additional initialization shock is present.

The gains rule is equivalently the complementarity system

$$\ell_t \geq 0, \qquad \ell_t \geq \tau^g(P_t - P_{t-1}), \qquad \ell_t[\ell_t - \tau^g(P_t - P_{t-1})] = 0. \tag{69}$$

It is continuous but kinked. Within an appreciating regime its price derivative is τ^g; within a non-appreciating regime it is zero.

Fix a terminal date J. From $J+1$ onward, hold prices and transfers at their new steady-state values. A household young at J evaluates its own old-age state at those values; because it bought at P_J, that transitional basis is retained when computing ℓ_{J+1}. Its continuation value is not replaced by a scalar. The state generated at $J+1$ is recorded rather than forced to equal the new-steady-state state. The resulting object is a terminally closed truncation, not necessarily an exact finite prefix of an infinite-horizon transition. Let z_J stack $(\log P_0, \ldots, \log P_J, \log T_0, \ldots, \log T_J)$ and order the residuals as the $J+1$ housing equations followed by the $J+1$ government equations. Recursively applying household policies, the distribution law, and cohort laws turns (26) into a finite system

$$\mathbf{R}_J(z_J; \varepsilon) = 0, \tag{70}$$

where ε scales the permanent change from ϑ^0 to ϑ^1 and the terminal steady state also varies with ε.

Because the constant path lies at the gains kink, local analysis is directional. Choose a candidate appreciation pattern $d = (d_0, \ldots, d_{J+1}) \in \{0,1\}^{J+2}$ and define the smooth branch

$$\ell^d_t = \tau^g d_t (P_t - P_{t-1}). \tag{71}$$

It is a branch of the statutory rule when $P_t - P_{t-1} > 0$ for $d_t = 1$ and $P_t - P_{t-1} < 0$ for $d_t = 0$. Let $\mathbf{R}^d_J$ denote the residual system evaluated on this branch.

Proposition 5 (Directional local existence of a truncation)**.** *Suppose the old steady state is regular, the household active sets are strict, and the branch Jacobian $\partial \mathbf{R}^d_J / \partial z_J$ is nonsingular at $\varepsilon = 0$. Suppose also that the directional solution has*

$$\left.\frac{\mathrm{d}(P_t - P_{t-1})}{\mathrm{d}\varepsilon}\right|_0 > 0 \text{ if } d_t = 1, \qquad \left.\frac{\mathrm{d}(P_t - P_{t-1})}{\mathrm{d}\varepsilon}\right|_0 < 0 \text{ if } d_t = 0. \tag{72}$$

Then, for each fixed J, there is a locally unique terminally closed truncation on the selected branch d for all sufficiently small positive shocks. It varies continuously differentiably with the shock, and satisfies the statutory gains rule on dates $0, \ldots, J+1$, including the continuation gain that enters the date-J household problem.

Proof. Steady-state regularity and Proposition 4 give a continuously differentiable terminal steady state for small ε. Strict active sets, positive log domains, the logit tenure share, and (71) make $\mathbf{R}_J^d$ continuously differentiable in (z_J, ε). At $\varepsilon = 0$, the constant old-steady-state path is a root on every branch. Nonsingularity and the implicit-function theorem give a unique differentiable branch solution. The strict directional signs in (72) persist for sufficiently small positive ε and validate every selected branch. Therefore $\ell_t^d = \tau^g \max\{P_t - P_{t-1}, 0\}$ and the branch solution solves the truncated statutory system. The argument does not rule out a solution on another self-consistent branch. □

If a directional price difference is zero, ordinary branch analysis is inconclusive at that date; higher-order or generalized-derivative methods are needed. Thus the proposition does not hide the steady-state kink inside a differentiability assumption.

Finite truncations do not by themselves establish existence or determinacy of an infinite-horizon transition. Such a result would require a separate stability argument for the equilibrium system or a proof that a sequence of truncated roots converges uniformly in the tail. The numerical construction below reports residual, terminal-state, and horizon comparisons only as diagnostics for the displayed finite-horizon object.

The reason one shock can affect several successive capital-gains bases is the demographic state. The initial fertility response changes Y_1 through $Y_1 = \nu \bar{n}_0 Y_0$; the resulting young cohort becomes the old cohort one period later, and subsequent fertility choices continue to update cohort masses. With a fixed housing stock, this propagation can move equilibrium housing demand and prices over several dates. Whenever a date in that path has $P_s > P_{s-1}$, the cohort selling at s realizes $\ell_s = \tau^g(P_s - P_{s-1})$. The construction below supplies the sign and duration for one parameterization rather than claiming that appreciation must occur after every fertility shock.

A.5 Conditional reallocation

For a constrained young owner, (51) gives the current-goods marginal value

$$MV_{it}^Y = \frac{\alpha x_{it}^O}{s_{it}^O} = q\, r_t + q\ell_{t+1} + \zeta_{it}^{O,F}. \tag{73}$$

For an old incumbent whose retention and estate-composition constraints are both slack, the housing first-order condition from (10) gives

$$MV_{jt}^O = \frac{\gamma c_{jt}^2}{h_{jt}^2} = q\, r_t - \ell_t. \tag{74}$$

Subtracting gives the decomposition in (29).

Assumption 1 (Two-date transfer ledger)**.** For an infinitesimal transfer of housing from incumbent j to buyer i, the allocation authority can transfer title and occupancy at date t, provide a pledgeable bridge loan at the bond price without changing other households' constraints, and make household-specific transfers at dates t and $t+1$. The buyer's next-period retention and estate-composition constraints are strictly slack. The authority holds the buyer's baseline old-age retained housing fixed and sells the added unit at $t+1$. It can return the two dates' incremental tax receipts while holding fertility, tenure, cohort masses, and all unaffected allocations fixed.

Proof of Proposition 2. Transfer $\epsilon > 0$ units from the old incumbent to the young buyer. The baseline has the incumbent retain the unit, transfer it at death with basis P_{t+1}, and have the estate sell it at date $t+1$ without gains tax. The following ledger values every change in date-t consumption goods and divides by ϵ:

Agent	Change relative to the baseline	Net value
Incumbent j	Receives P_t, pays ℓ_t, and gives up a unit whose net service-and-estate value, after property tax, is $P_t - \ell_t$ at the interior retention choice.	0
Buyer i	Pays P_t using market-rate bridge finance, pays $q\tau^p P_t$, receives current services worth MV_{it}^Y, and the added unit is sold next period for $q(P_{t+1} - \ell_{t+1})$.	$\zeta_{it}^{O,F}$
Government	Property-tax revenue is unchanged; the incumbent sale and the buyer's later sale yield revenue absent from the baseline estate sale.	$\ell_t + q\ell_{t+1}$
Lenders	Bridge and mortgage funds are supplied at the bond price.	0
Real resources	Transferring title and occupancy uses K_{ijt} units of current goods.	$-K_{ijt}$
Total		$\zeta_{it}^{O,F} + \ell_t + q\ell_{t+1} - K_{ijt}$

For the buyer row, substituting $MV_{it}^Y = q\,r_t + q\ell_{t+1} + \zeta_{it}^{O,F}$ and $q\,r_t = P_t + q\tau^p P_t - qP_{t+1}$ gives the displayed net value. For the incumbent row, the retention first-order condition makes the marginal unit's net service-and-estate value equal to its after-tax sale proceeds: $MV_{jt}^O - q\tau^p P_t + qP_{t+1} = P_t - \ell_t$. Hence no occupancy ledger is being combined with the title-transfer ledger.

The buyer need not choose to resell one additional unit after unrestricted old-age reoptimization. Assumption 1 instead holds the buyer's baseline old-age retained housing fixed and assigns the added unit to the date-$(t+1)$ sale. Strict slackness makes this perturbation feasible, and the old-age first-order conditions make the restriction welfare-neutral to first order.

If total net value is positive, date-t and date-$(t+1)$ transfers can leave both affected households weakly above baseline and one strictly above it. Smoothness and the strict first-order inequality imply the same conclusion for a sufficiently small finite transfer. At a steady state, $\ell_t = \ell_{t+1} = 0$, so the condition becomes $\zeta_i^{O,F} > K_{ij}$. □

The transfer ledger is part of the proposition. Without pledgeable bridge finance or two-date transfers, the marginal-value gap remains a shadow-value decomposition but need not be implementable as a Pareto improvement. The steady-state term $\zeta_i^{O,F}$ is the shadow value of relaxing an exogenous financing constraint; the model does not provide a deeper microfoundation for that constraint. The proposition also fixes population. Policies that change fertility compare economies with different sets of people, for which a social criterion must be stated explicitly; see Golosov et al. (2007) and Pérez-Nievas et al. (2019). The paper therefore reports the conditional allocation result rather than a global welfare theorem.

The tax treatment of rental intermediaries affects incidence but not the decomposition. If an intermediary also pays ℓ_{t+1} upon resale, its zero-profit rent is $r_t^L = r_t + \ell_{t+1}$. The young owner's private current-goods cost becomes $q\,r_t^L$, while the incumbent's retention cost can be written $q\,r_t^L - \ell_t - q\ell_{t+1}$. Their difference remains $\zeta_{it}^{O,F} + \ell_t + q\ell_{t+1}$.

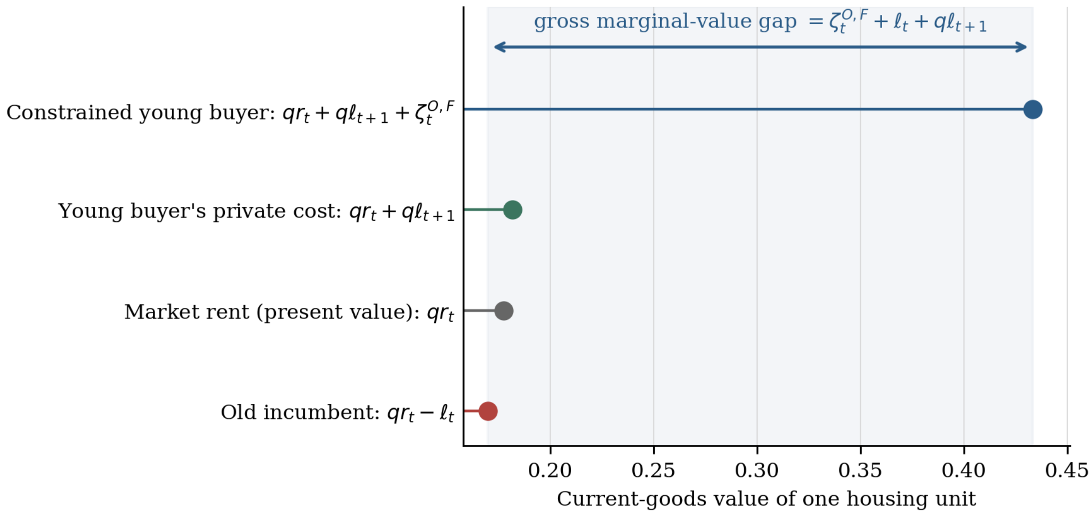

Figure 2: Shadow-value decomposition on the computed path. The financing wedge raises the young buyer's value; realization taxes lower the old incumbent's value and raise the buyer's anticipated private cost. The displayed object is the gross marginal-value gap. Under the ledger in Proposition 2, a compensated improvement obtains when this gap exceeds the real transfer cost.

A.6 Housing access and fertility

Suppose $h = \widehat{h}$ binds and fertility remains interior. Define

$$x = \frac{\widetilde{w}_{it}^m - u_t^m \widehat{h} - \chi n}{1 + k_{t+1}^m}, \qquad s = \widehat{h} - \kappa n, \tag{75}$$

and collect the positive terms in

$$\Delta = \frac{\vartheta_t}{n^2} + \frac{\chi^2}{(1 + k_{t+1}^m)x^2} + \frac{\alpha\kappa^2}{s^2} > 0. \tag{76}$$

Implicit differentiation of the binding-fertility condition gives

$$\frac{\mathrm{d}n}{\mathrm{d}\widehat{h}} = \frac{\alpha\kappa/s^2 - \chi u_t^m/[(1 + k_{t+1}^m)x^2]}{\Delta}. \tag{77}$$

Derivation of (77) *and* (30). Let the left side of (53) be $F(n, \widehat{h})$. Direct differentiation gives

$$F_n = -\Delta < 0, \tag{78}$$

$$F_{\widehat{h}} = \frac{\alpha\kappa}{s^2} - \frac{\chi u_t^m}{(1 + k_{t+1}^m)x^2}. \tag{79}$$

The implicit-function theorem gives $\mathrm{d}n/\mathrm{d}\widehat{h} = -F_{\widehat{h}}/F_n$, which is (77). At a constrained allocation, (51) implies

$$\frac{x}{s} = \frac{u_t^m + \zeta_{it}^m}{\alpha}. \tag{80}$$

Substituting this ratio into $F_{\widehat{h}} \geq 0$ gives the condition in (30) on the interior old-age branch, where $k_{t+1}^m = \beta A$. On the capped old-renter branch the same derivation uses $k_{t+1}^R = \beta(1 + \omega_B)$. □

The same calculation separates two other partial-equilibrium channels:

$$\frac{\partial n}{\partial \widetilde{w}_{it}^{m}} = \frac{\chi}{(1+k_{t+1}^{m})x^2\Delta} > 0, \qquad \frac{\partial n}{\partial u_t^m} = -\frac{\chi \widehat{h}}{(1+k_{t+1}^{m})x^2\Delta} < 0. \tag{81}$$

At a fixed cap, more lifetime resources raise fertility and a higher housing user cost lowers it.

A primitive sufficient condition, stronger than the exact condition, is

$$\chi \leq \frac{(1+k_{t+1}^{m})\kappa u_t^m}{\alpha}, \tag{82}$$

because $\zeta_{it}^{m} \geq 0$. When the down-payment constraint alone binds,

$$\widehat{h}_{it}^{O} = \frac{b_i}{(1-\phi)P_t}, \qquad \frac{\partial \widehat{h}^{O}}{\partial b_i} = \frac{1}{(1-\phi)P_t}, \qquad \frac{\partial \widehat{h}^{O}}{\partial \phi} = \frac{b_i}{(1-\phi)^2 P_t}. \tag{83}$$

The derivative with respect to ϕ isolates the cap-relaxation channel when prices and resources are held fixed. The two total derivatives on this branch are

$$\frac{\partial n}{\partial \phi} = \frac{\partial n}{\partial \widehat{h}} \frac{b_i}{(1-\phi)^2 P_t}, \tag{84}$$

$$\frac{\partial n}{\partial b_i} = \frac{\partial n}{\partial \widetilde{w}_{it}^{O}} + \frac{\partial n}{\partial \widehat{h}} \frac{1}{(1-\phi)P_t}. \tag{85}$$

A change in liquid wealth has both a wealth channel and a housing-access channel. A funded policy additionally moves prices, transfers, and tenure, and its aggregate response requires the equilibrium system.

A.7 Solved mixed-tenure transition approximation

The numerical construction specializes observed types to common income and liquid wealth while retaining the logistic ownership-preference heterogeneity in (11). It computes steady-state equilibria and a terminally closed transition approximation; it is neither a calibration nor an infinite-horizon existence or determinacy result. Table 1 lists the parameters.

Table 1: Parameters of the mixed-tenure construction

Object	Parameter	Value	Object	Parameter	Value
Bond price	q	0.50	Mortgage share	ϕ	0.80
Property tax	τ^p	0.05	Gains-tax rate	τ^g	0.20
Space weight	α	0.40	Space per child	κ	0.50
Goods per child	χ	0.15	Young discount	β	0.40
Old housing weight	γ	0.30	Estate weight	ω_B	0.40
Income	y	1.00	Liquid wealth	b	0.06
Household formation	ν	2.00	Housing stock	$\bar{H}$	2.00
Rental limit	$h_R^{\max}$	0.45	Owner limit	$h_O^{\max}$	2.00
Owner intercept	$\bar{\xi}$	−0.15	Taste scale	σ_ξ	0.12

At date zero, ϑ rises permanently from 0.35 to 0.50. Solving (24) gives Table 2.

Average fertility is $1/\nu = 0.5$ at both endpoints, but conditional fertility, tenure, housing, prices, and population scale differ. The determinants of the two steady-state Jacobians, computed in

Table 2: Positive steady-state endpoints

Object	Initial steady state	Final steady state
Fertility preference ϑ	0.3500	0.5000
Average fertility $\bar{n}$	0.5000	0.5000
House price P	0.3427	0.4844
Per-household transfer T	0.00589	0.00602
Young owner share	0.7486	0.5149
Renter fertility	0.3527	0.4368
Owner fertility	0.5495	0.5596
Young renter housing	0.4500	0.4500
Young owner housing	0.8754	0.6193
Old renter housing	0.4500	0.4500
Old owner housing	0.6582	0.4649
Young and old cohort mass	1.4553	2.0103

$(\log P, \log T)$, are -0.00466 and -0.00433, so both roots are locally regular. Their maximum equation residuals are below 2×10^{-11}.

As a diagnostic for the sufficient conditions in Proposition 4, a 41×31 grid over $P \in [0.25, 0.60]$ and $T \in [0.005, 0.020]$ keeps the fiscal map inside the transfer interval for both values of ϑ. The largest centered finite-difference value of $|\Psi_T|$ is 0.00588. At the fiscal fixed points on the two price faces, the replacement residuals are $(0.253, -0.291)$ initially and $(0.687, -0.108)$ finally. These are reproducible numerical checks of the stated directions on the declared grid, not interval-arithmetic proofs over the continuum.

The horizon-28 truncation holds prices and transfers at their final steady-state values from date 29 onward and solves all date-by-date housing and government equations through date 28 jointly. A household young at date 28 evaluates its own old-age state in that stationary environment. The state generated at date 29 is recorded rather than imposed. On impact, average fertility rises to 0.6223; the young cohort then expands with the OLG lag. The price moves from 0.3427 before the shock to 0.3800 on impact, continues to 0.4975 by date 3, and then approaches 0.4844 with damped oscillations. Realized gains taxes exceed 10^{-4} per unit on dates 0–3 and, after the initial overshoot, again on dates 7–8. Smaller oscillations continue in the numerical tail.

The computed truncation has both tenures on every displayed date: the owner share lies between 0.4830 and 0.7900. Minimum renter and owner saving are 0.3517 and 0.4070. The unconstrained young renter and owner housing policies exceed their respective limits by at least 0.5337 and 0.3893, so both young limits bind strictly. The owner-size limit has slack of at least 1.2106, so the owner wedge is purely financial. The old renter cap has a minimum shadow wedge of 0.02057. Old owners sell at least 0.1472 units, their incumbent user cost is at least 0.1699, and their estate-composition slack is at least 0.00990; hence the old-owner interior formulas apply strictly throughout.

The maximum absolute scaled housing or government residual is 6.51×10^{-10}. The maximum terminal-state discrepancy is 1.88×10^{-9}. Replacing horizon 28 by horizon 24 changes the first nine dates in prices, transfers, masses, tenure, fertility, and housing by at most 1.78×10^{-15}. At the computed horizon-28 root, freezing the realized appreciation pattern gives a smallest stacked-Jacobian singular value of 0.350 and a condition number of 56.5. These are conditioning diagnostics for that finite root, not a verification of the directional derivative assumptions at a zero-size shock. Finally, the analytical fertility derivative is 0.6514544143; a centered finite difference gives 0.6514544143, with absolute error 3.3×10^{-12}.

Longer terminal horizons give a second check on the tail:

Terminal date J	Maximum terminal-state gap
20	3.86×10^{-7}
24	2.75×10^{-8}
28	1.87×10^{-9}
32	9.73×10^{-11}
36	2.93×10^{-11}
40	1.88×10^{-11}

The first twenty dates of the horizon-28 and horizon-32 solutions differ by at most 3.11×10^{-15}.

The specialized economy also permits a local saddle-path diagnostic. Its seven inherited variables are the two cohort masses, the current old's owner share, renter and owner financial wealth, inherited owner housing, and the lagged price basis. Setting $Q_t = P_{t+1}$ writes the system with two forward price coordinates, P_t and Q_t; the government budget locally determines T_t. The ten-dimensional descriptor pencil has seven stable finite roots, two unstable finite roots, and one infinite generalized eigenvalue from the static fiscal equation. Eliminating T_t leaves a regular nine-dimensional system with the same finite roots, giving the saddle-path configuration. The table reports the final-steady-state roots for each one-sided derivative of the current and next positive-part gains terms:

d_t	d_{t+1}	Stable	Finite unstable	$\max \lvert\lambda_s\rvert$	$\min \lvert\lambda_u\rvert$
0	0	7	2	0.5465	3.8919
0	1	7	2	0.5332	3.9322
1	0	7	2	0.4945	3.7456
1	1	7	2	0.4786	3.7830

No finite root lies on the unit circle, and each stable subspace projects with full rank onto the seven predetermined states. All 441 points of a 21×21 grid over independently selected gains slopes $(d_t, d_{t+1}) \in [0,1]^2$ have the same count. Across the grid, the largest stable modulus is 0.5465, the smallest unstable modulus is 3.7456, and the smallest distance from the unit circle is 0.4535. The dominant stable roots are complex, consistent with the damped oscillations. These calculations are local saddle-path evidence, not a nonsmooth stable-manifold proof of infinite-horizon existence or determinacy at the gains-tax kink.

The original young budgets, fertility conditions, and saving Euler equations are evaluated after the branch-specific reductions. Across both tenures and all transition dates, the largest absolute normalized Euler residual is 2.4×10^{-16}, the largest budget residual is 2.3×10^{-16}, and the largest fertility first-order-condition residual is 3.4×10^{-16}. The largest residual in the full price-of-a-child identity (20) is 2.3×10^{-16}. The capped old-renter branch and interior old-owner branch are selected on every date.

The construction is generated by `code/model/tools/build_simplified_olg_mixed_tenure_theory.py`. It writes the steady states, full truncated path, verification ledger, and the two figures used in the paper. These files make the finite-root, household-optimality, and active-set claims reproducible; they do not establish an infinite-horizon transition. The descriptor calculation is generated by `code/model/tools/build_simplified_olg_saddle_path_diagnostic.py`.

References

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